



ML3 PROJECT PRESENTATION

TEAM 12:

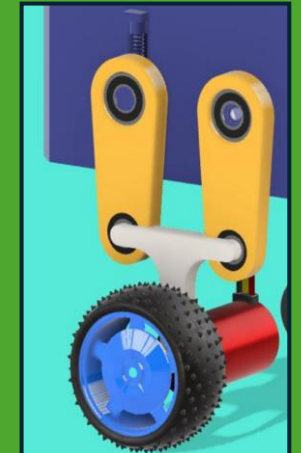
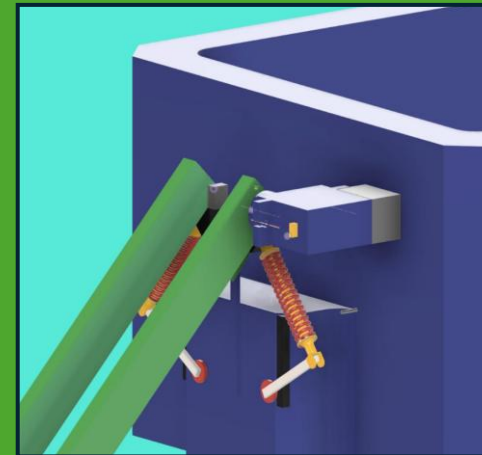
Mohnish, Abdullah, Daphne, Catherine and Ayaat



Robot Concept : (Tri-wheel)

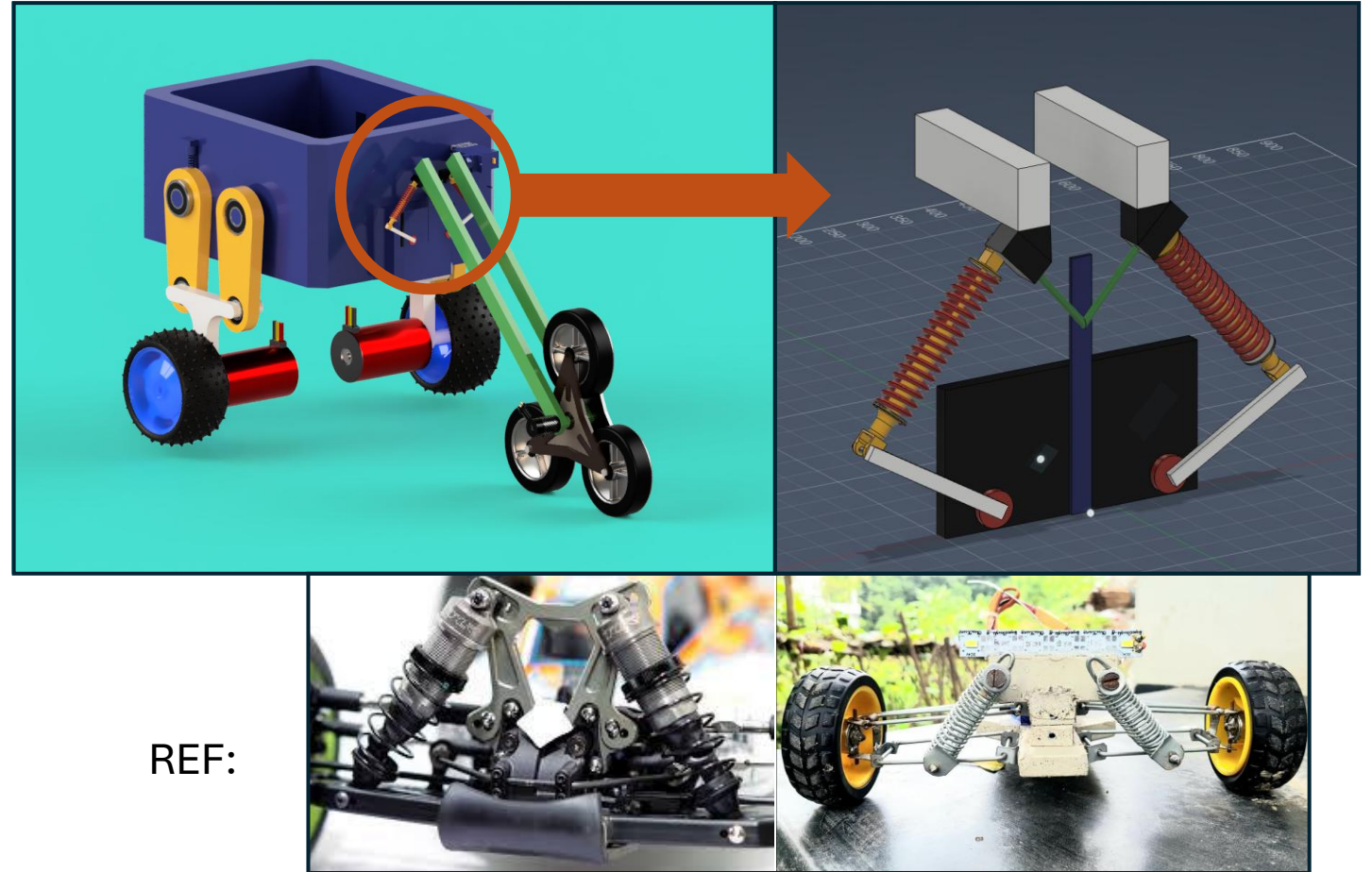
The Tri-wheel can be divided into 3 separate parts

- Front Half
- Rear Half
- Top Half



Front Half Spring Mechanism:

- The suspension helps reduce stress on the front wheel
- Assist with the drop
- The front wheel is the main driving force of the robot



Front Half

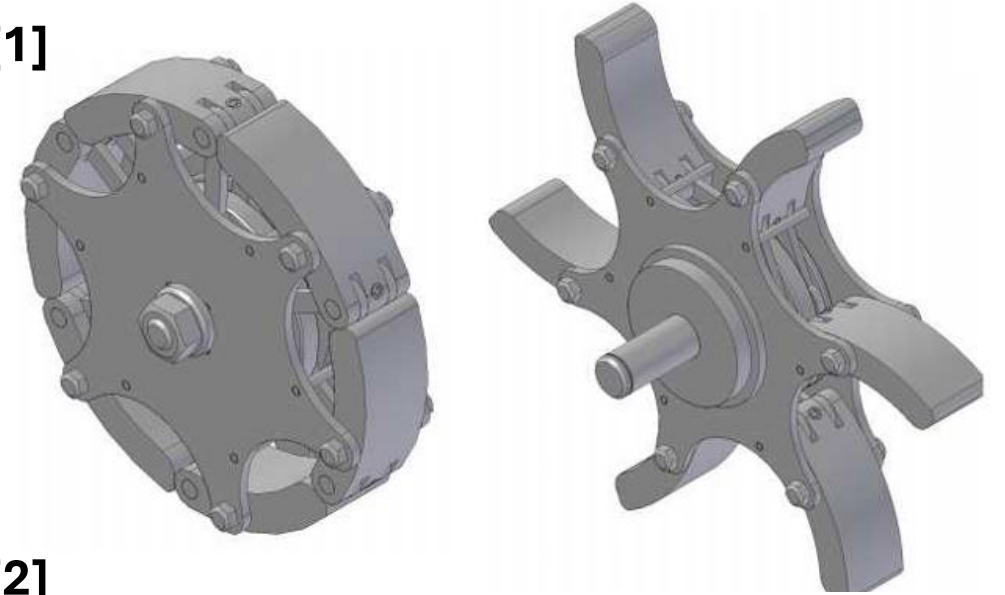


Configurable wheel that has two forms closed and open configurations

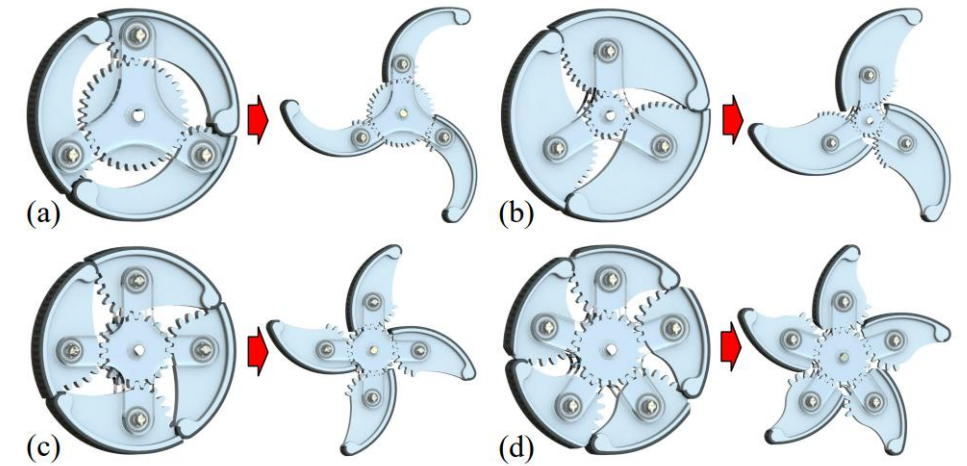


With the help of a gears, we can dynamically switch between the two configurations

[1]



[2]

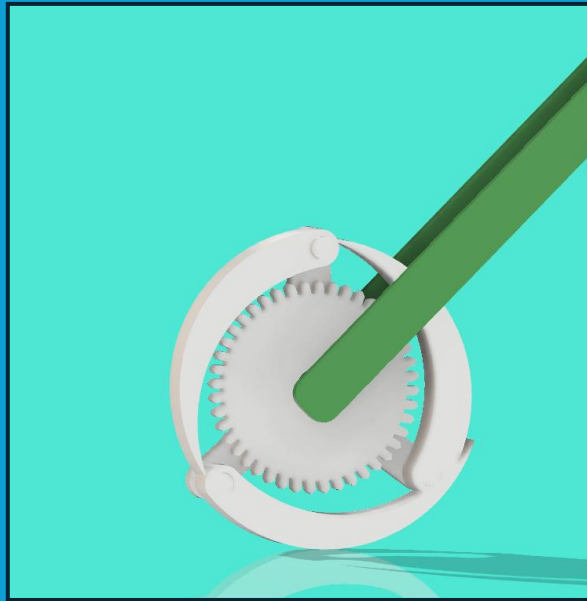


[1] R. Sell, G. Aryassov, A. Petritshenko, and M. Kaeeli, "Kinematics and dynamics of configurable wheel-leg," in *Proc. 8th Int. DAAAM Baltic Conf. Industrial Engineering*

[2] C. Zheng and K. Lee, "Wheeler: Wheel-leg reconfigurable mechanism with passive gears for mobile robot applications," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*,

Front Half

IDEA1

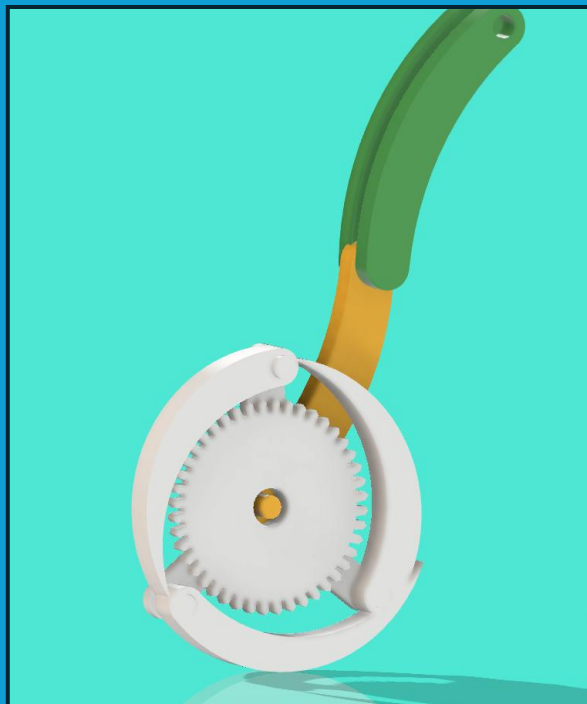


The wheel is connected to the main body of the robot via an extended rod

Idea 1 is a simpler and easily manufacturable version

Idea 2 has an added link to decrease shock

IDEA 2



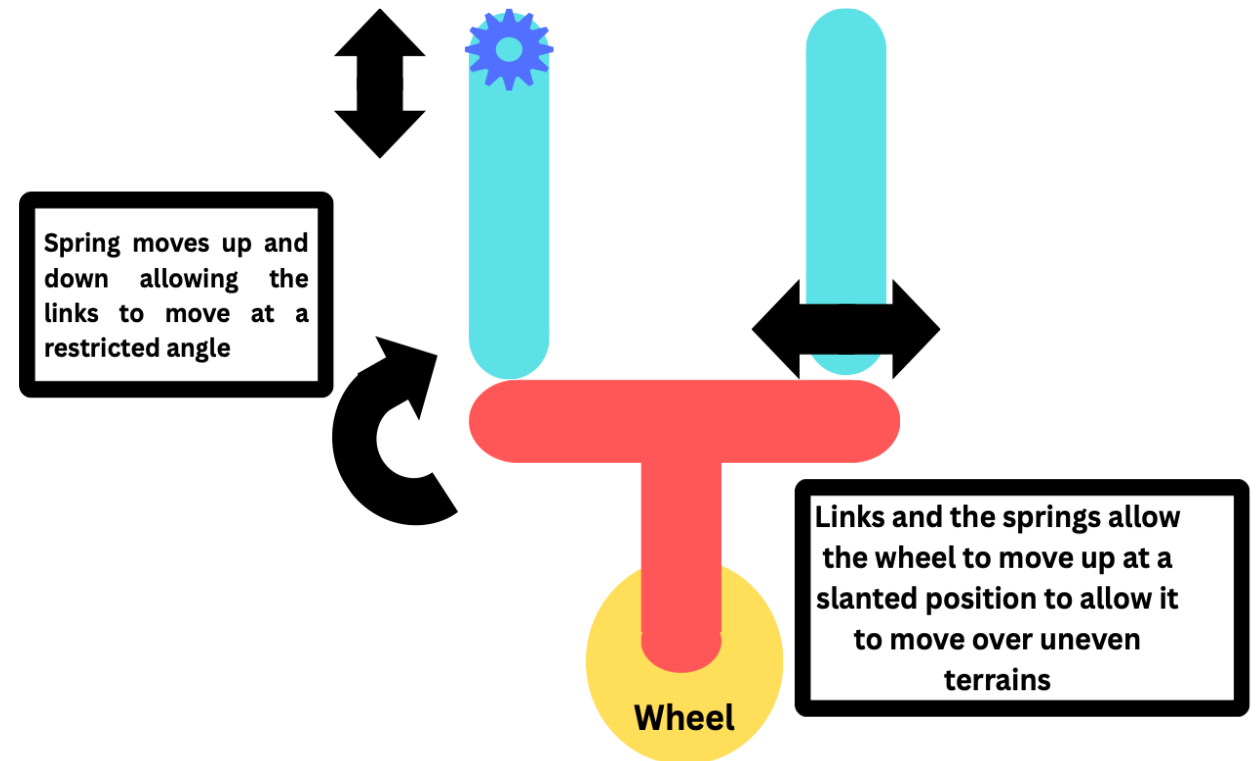
The front rod helps with:

- Stairs
- Drop
- Ditch



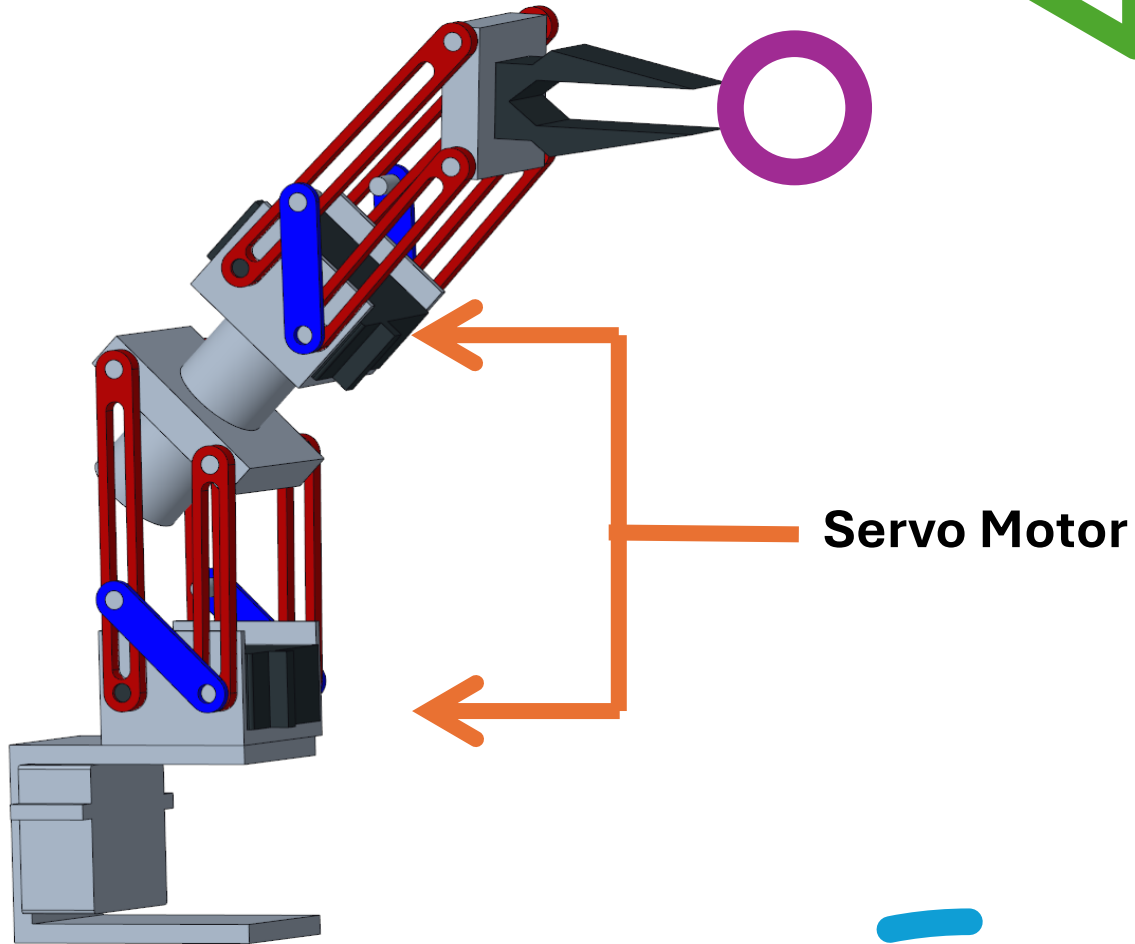
Rear Half

The Rear half consists of two generic wheels that are connected to a 4-bar spring



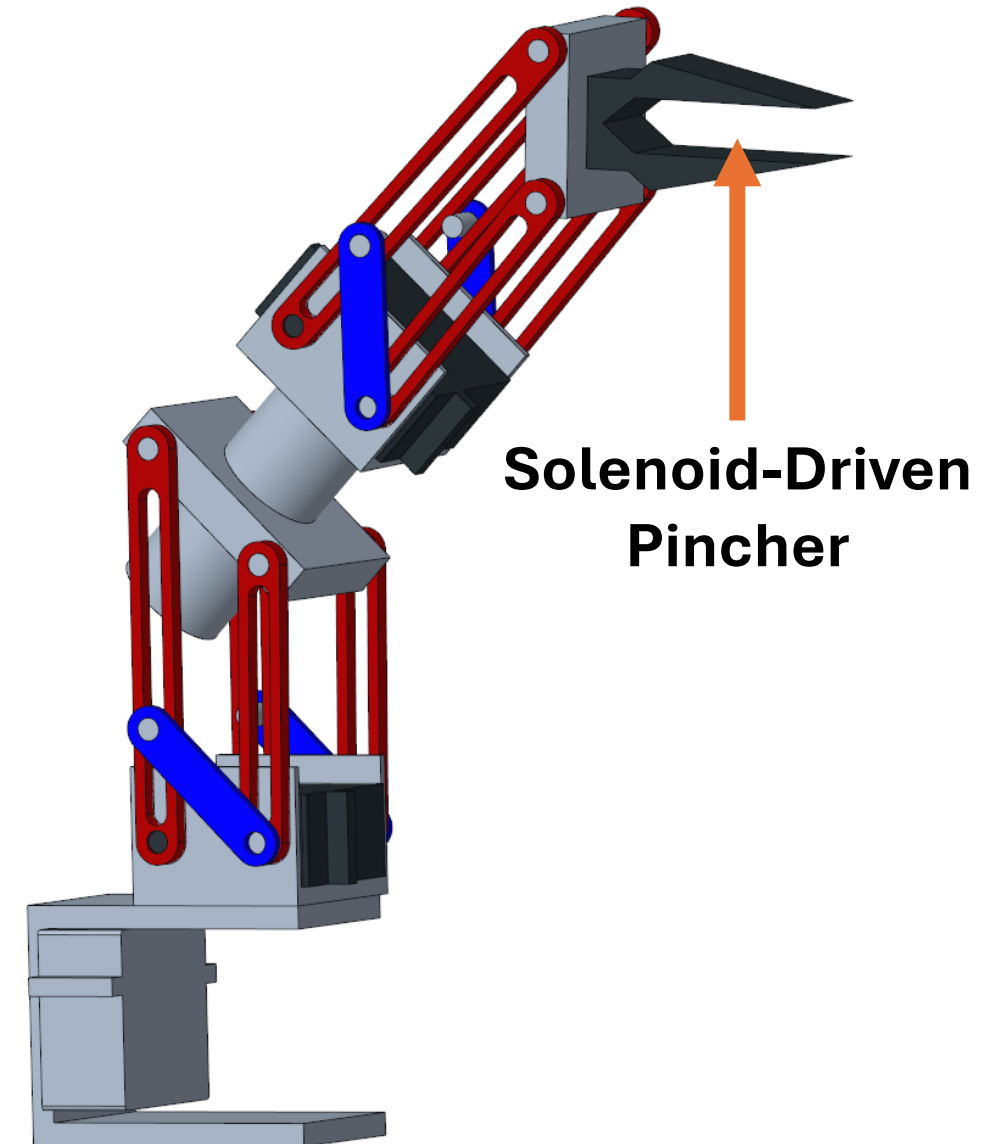
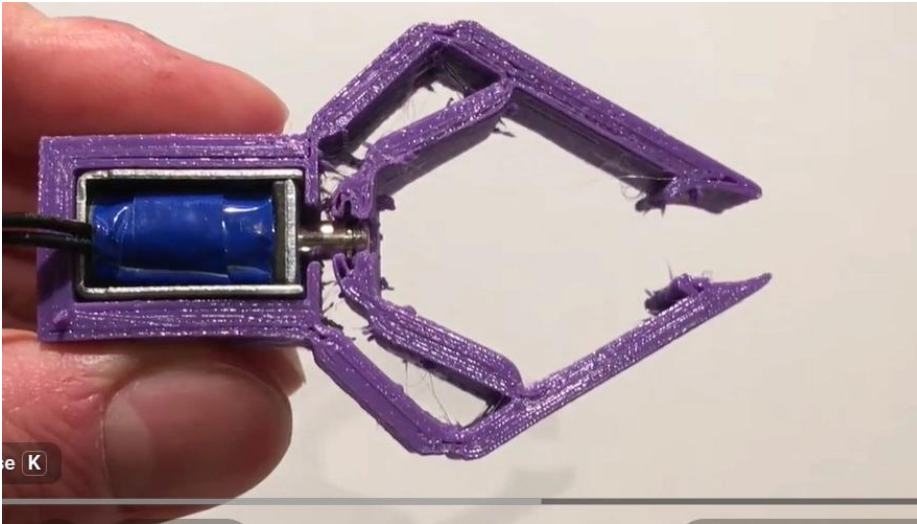
Top Half

The 2R-manipulator consists of 2 servo motors that provide us with 1 DoF in the forward direction



Top Half

- With the help of a **solenoid actuator**, we can open and close the pincher of the manipulator



Originality

- The use of a Tri-wheel design to be able to overcome the ditch as well as the ledge.
- Incorporate a configurable wheel-leg to aid in stair climbing.
- Implementation of spring-based support to act as suspension for uneven terrain

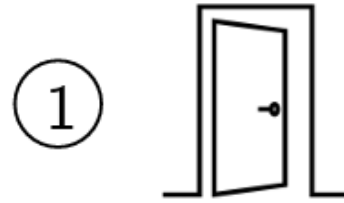


Motivation

- To further develop engineering and fabrication skills
- To gain hands-on experience working with all-terrain robotic platforms
- To understand how robotic design can help overcome challenging obstacles

Approach for Stage-I

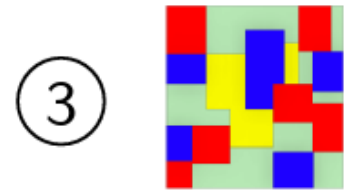
The robot will push the door



Spring suspension at the back wheel



Configurable Wheel



Scanning using camera



Spring suspension at the back wheel

⑥



Configurable Wheel

⑦



⑧



Using camera

⑨



2R manipulator with Solenoid-Driven Pincher

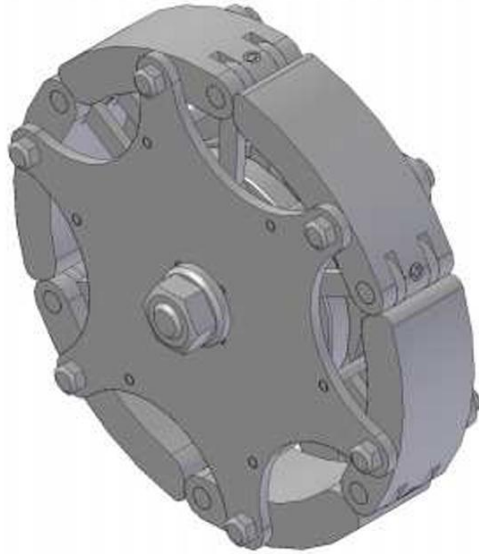
⑩



Approach for Stage-I (Configurable Wheel)

Closed Configuration

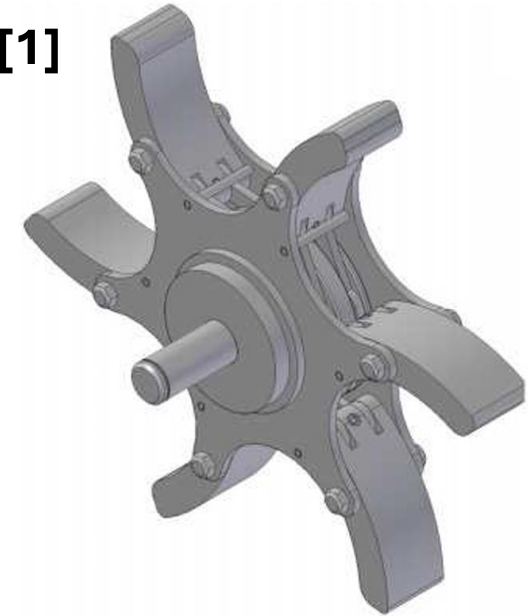
[1]



- Functional in smooth terrain
- Enables us to overcome the **bump** and **slope** (obstacle 2 and 6 respectively)

Open Configuration

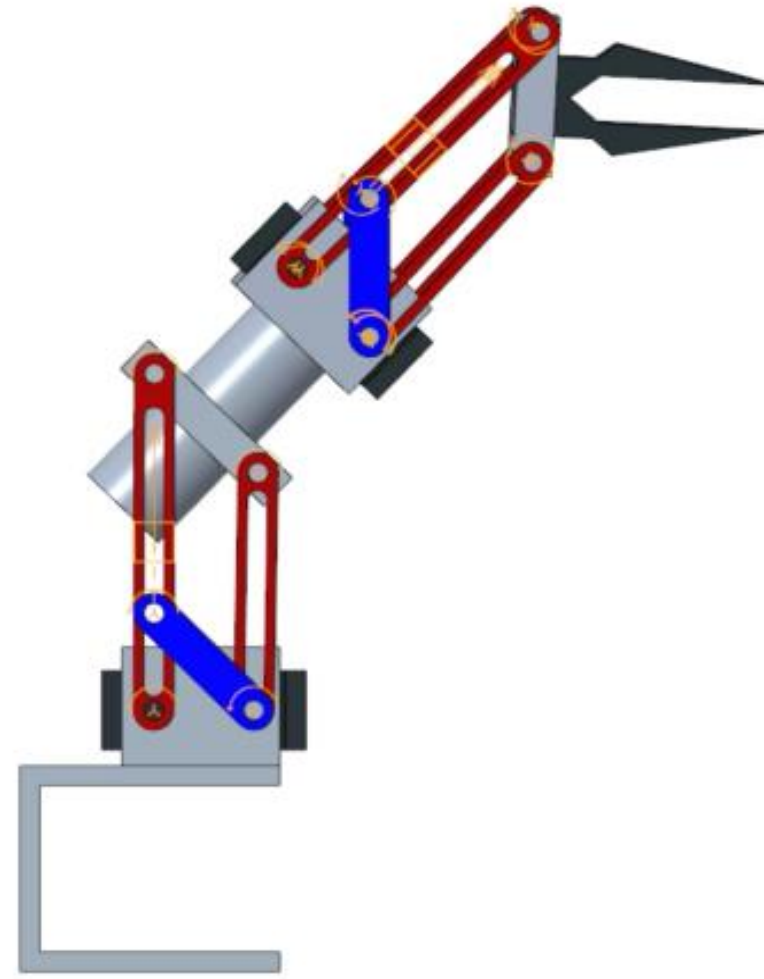
[1]



- Functional in uneven terrain
- Enables us to overcome the **ditch**, **roller** and **stairs** (obstacle 4, 7, and 8 respectively)

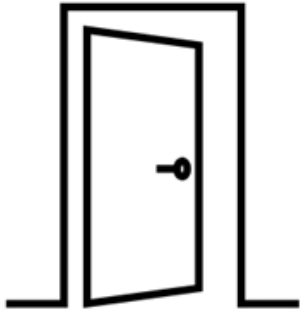
Approach for Stage-I 2R Manipulator

- The 2R manipulator will be able to push
- Solenoid-Driven Pincher will enable us to **pick up the block**



Approach for Stage-II

①



The door must be
opened not pushed

⑤



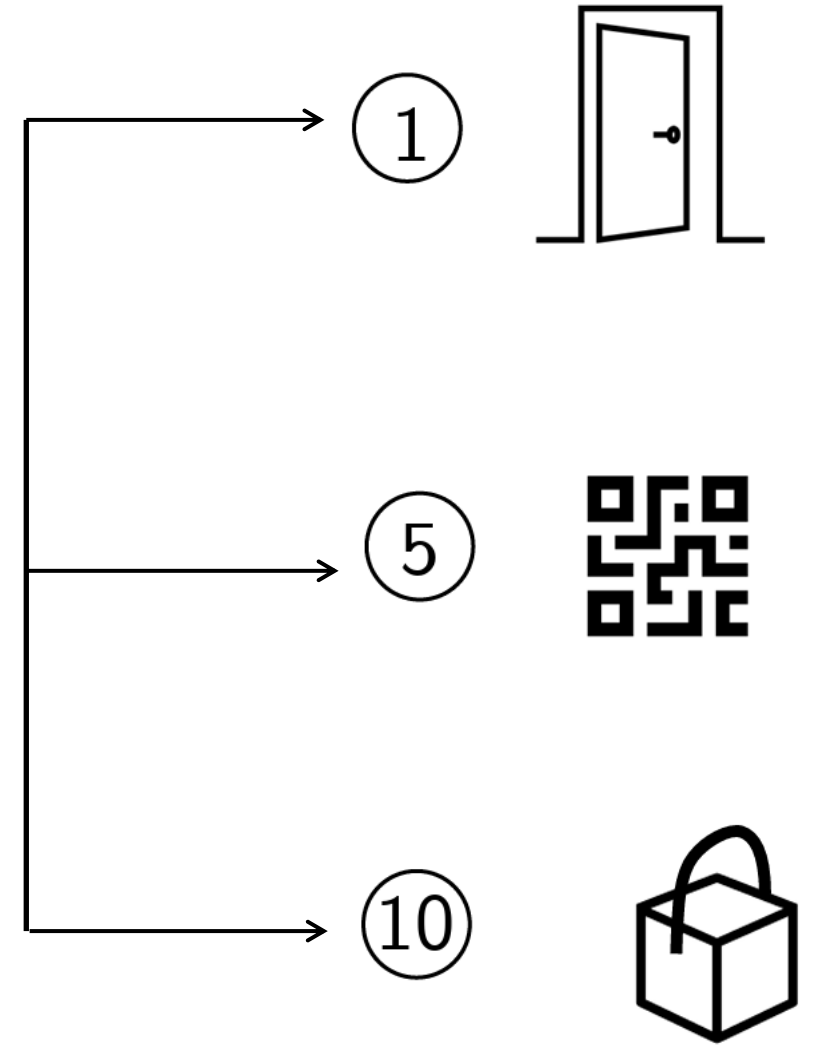
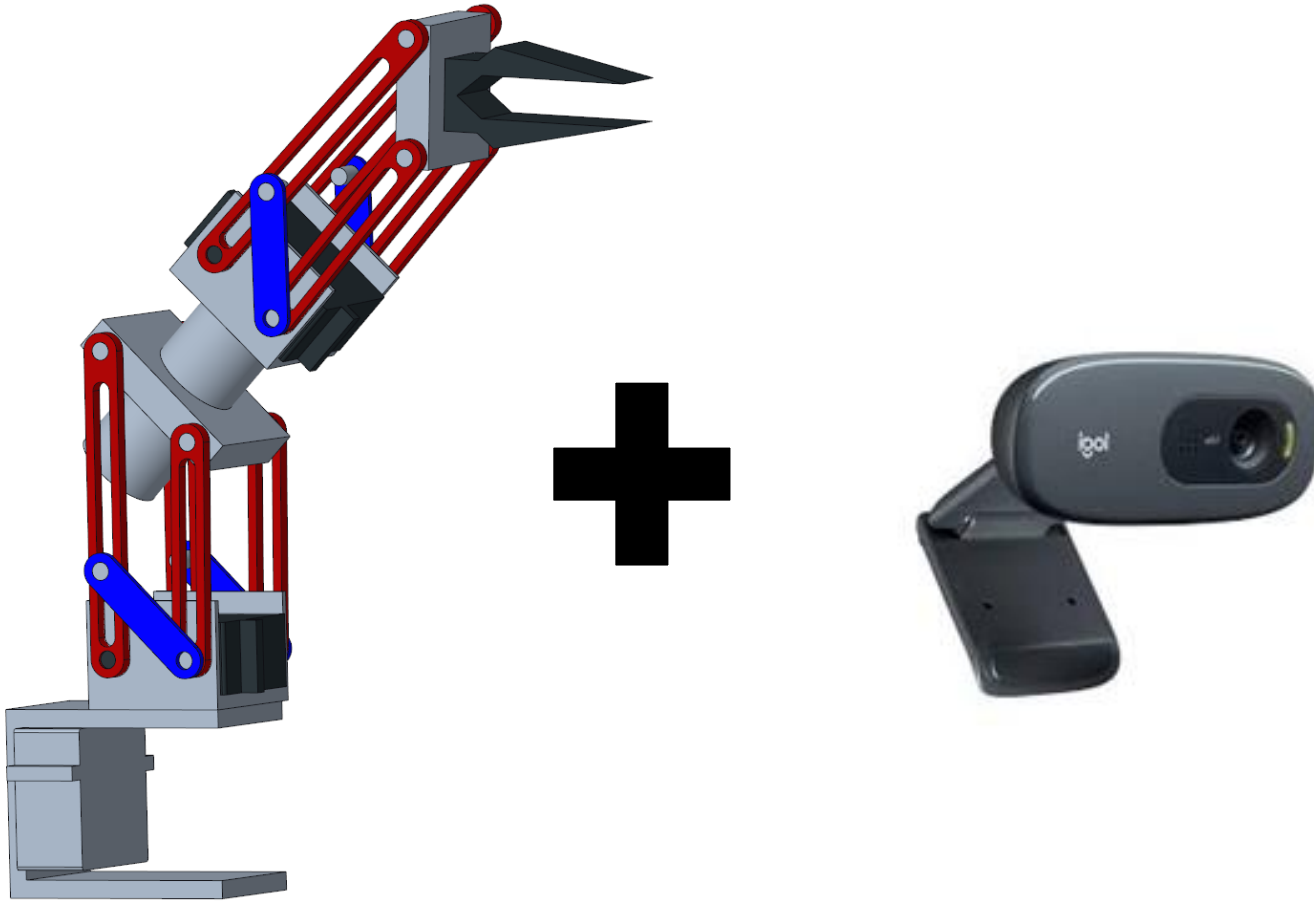
The QR code
must be **flipped**

⑩



The Block must
be **picked**

Approach for Stage-II



Approach for Stage-II

Approach for Obstacle 1 (Door Opening)

1

Detect door knob (YOLO + OpenCV)

2

Align gripper to knob in image space

3

Advance in small steps, re-verify each time

4

Proximity sensor halts forward motion

5

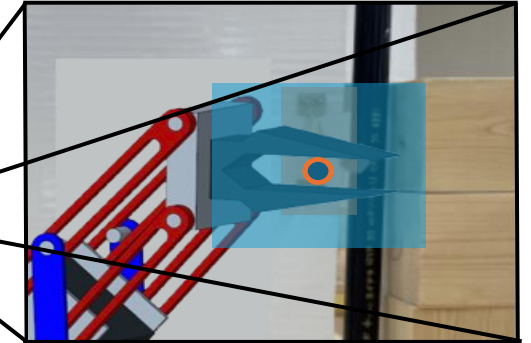
Close gripper — contact sensor confirms grasp

6

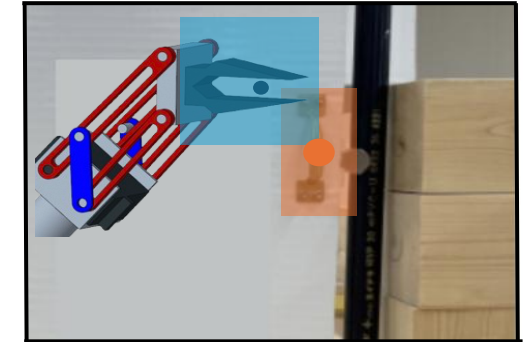
Pull knob — camera verifies door moved



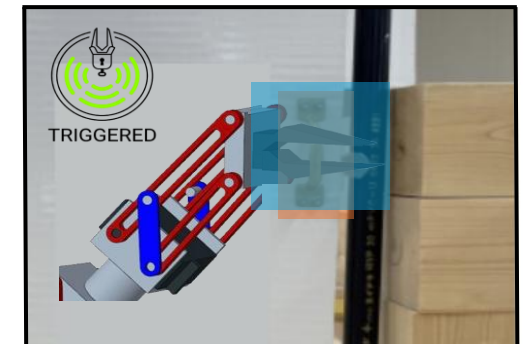
RGB Camera detects knob & gripper



Visual Servoing minimizes image-space position error



Proximity Sensor triggers grasp sequence precisely

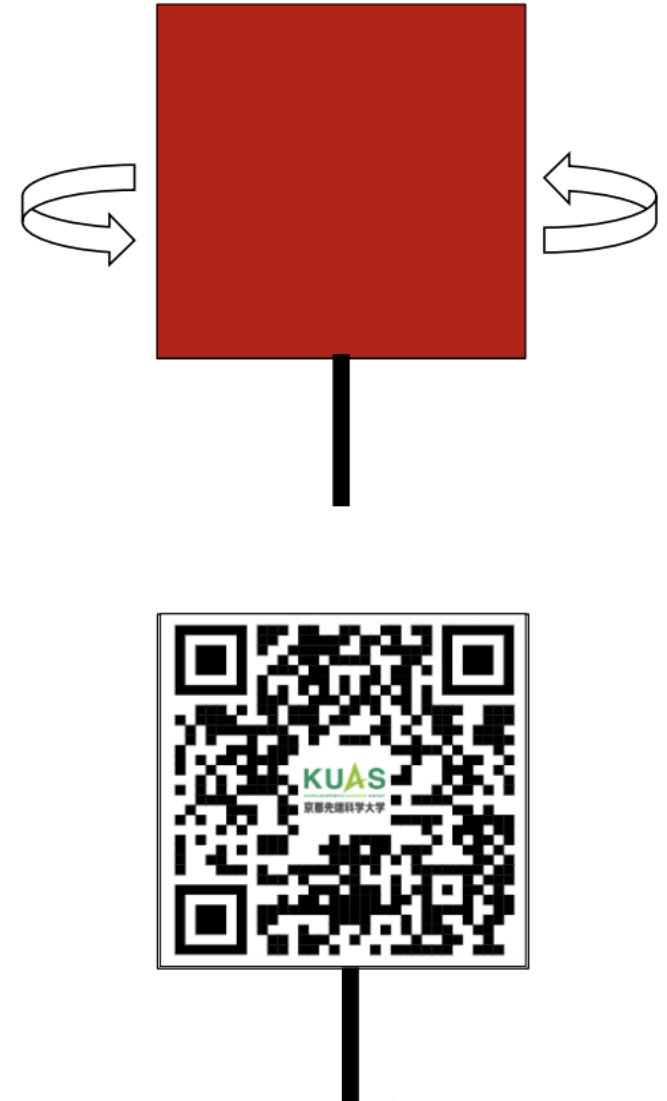


Contact Switch confirms successful grip on knob

Approach for Stage-II

Approach for Obstacle 5 (QR Code)

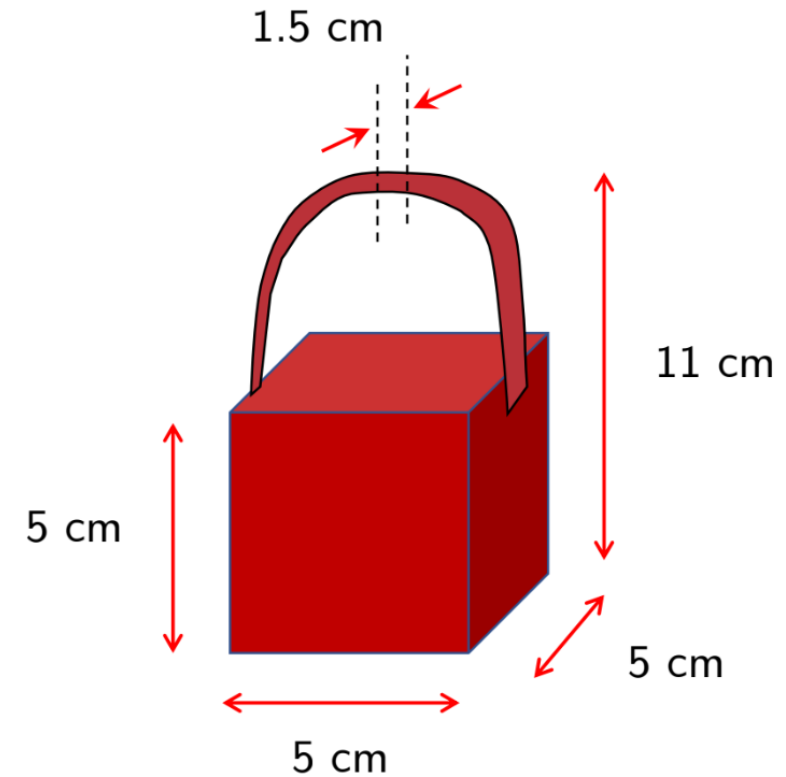
- Detect red cardboard using RGB camera
- Move end-effector toward the side of the cardboard using the same manipulation strategy as Obstacle 1
- Push and rotate the cardboard
- Scan the QR code using the onboard camera



Approach for Stage-II

Approach for Obstacle 10 (Block Picking)

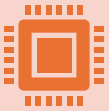
- Use teleoperation control for robot and manipulator movement
- Stream live camera feed from the robot to the control station
- Operator visually guides the manipulator to align with and grasp the block
- Carry the block safely to the target location



Role Assignment



Robot Design and Fabrication- All Members



Software and Control – Abdullah, Mohnish



Circuit making – Ayaat, Catherine, Daphne



Scheduling, E-Portal, and Progress Tracking - Daphne

List of Components and Estimated Budget

No.	Component	Quantity	Use	Price
1	Servo Motor	4	Arm/Locomotion	IN box
2	BLDC Motor	3	Locomotion	IN box
3	Motor driver	4	Control	IN box
4	Wheels	3	Locomotion	Printable
5	3D Filament	2	Body	¥5,000
6	Screws	TBD	Whole robot	¥2,000
7	Solenoid-actuator	1	Pincher	¥1,500
8	Suspension Mechanism	1	For the front wheel	¥1500
Total				¥10,000

Timeline for Robot Development

	WEEK																
TASK	7		8		9		10		11		12		13		14		15
	5/26	5/29	6/2	6/5	6/9	6/12	6/16	6/19	6/23	6/26	6/30	7/3	7/7	7/10	7/14	7/17	7/24
Design development																	
Parts manufacturing																	
Prototyping																	
Further development																	
Preparation and test runs																	
Presentation																	

Thank You

Q&A